EXERCISE 9.1

1. Find an approximate value of $\int_1^{1.5} x\,dx$ by applying the left-end rule with the partition $\{1.1, 1.2, 1.3, 1.4, 1.5\}$.
2. Find an approximate value of $\int_1^{1.5} x^2\,dx$ by applying the right-end rule with the partition $\{1.1, 1.2, 1.3, 1.4, 1.5\}$.
3. Find an approximate value of $\int_1^{1.5} (2-x)\,dx$ by applying the mid-point rule with the partition $\{1.1, 1.2, 1.3, 1.4, 1.5\}$.

9.2.2 Limit Formula to Evaluate $\int_a^b f(x)\,dx$

Divide the interval $[a,b]$ into n equal subintervals $[x_0,x_1],[x_1,x_2],\cdots,[x_{n-2},x_{n-1}],[x_{n-1},x_n]$ such that $a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b$. Then, we have $x_1 - x_0 = x_2 - x_1 = \cdots = x_n - x_{n-1} = \frac{b-a}{n}$. Put $h = \frac{b-a}{n}$. Then, we get $x_i = a + ih, i = 1, 2, \cdots, n.$

So, by the definition of definite integral, we get

$$\lim_{n\to\infty \text{ and } \max(x_i - x_{i-1})\to 0} \sum_{i=1}^{n} f(x_i)(x_i - x_{i-1}) \text{ (Right-end rule)}$$

$$= \lim_{n\to\infty} \frac{b-a}{n} \sum_{i=1}^{n} f\left(a + i\frac{b-a}{n}\right).$$

$$\therefore \quad \int_a^b f(x)\,dx = \lim_{n\to\infty} \frac{b-a}{n} \sum_{r=1}^{n} f\left(a + (b-a)\frac{r}{n}\right).$$

Note. $\lim_{n\to\infty} \frac{b-a}{n} \sum_{r=0}^{n} f\left(a + (b-a)\frac{r}{n}\right) = \lim_{n\to\infty} \left[\frac{b-a}{n} f(a) + \frac{b-a}{n} \sum_{r=1}^{n} f\left(a + (b-a)\frac{r}{n}\right)\right]$

$$= \lim_{n\to\infty} \frac{b-a}{n} \sum_{r=1}^{n} f\left(a + (b-a)\frac{r}{n}\right)$$

$$= \int_a^b f(x)\,dx.$$

$$\therefore \int_a^b f(x)\,dx = \lim_{n\to\infty} \frac{b-a}{n} \sum_{r=1}^{n} f\left(a + (b-a)\frac{r}{n}\right)$$

$$= \lim_{n\to\infty} \frac{b-a}{n} \sum_{r=0}^{n} f\left(a + (b-a)\frac{r}{n}\right).$$

If $a = 0$ and $b = 1$, then we get $\int_0^1 f(x)\,dx = \lim_{n\to\infty} \frac{1}{n} \sum_{r=0}^{n} f\left(\frac{r}{n}\right) = \lim_{n\to\infty} \frac{1}{n} \sum_{r=1}^{n} f\left(\frac{r}{n}\right).$

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